

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2025

Course: B. Tech

Branch: Civil Engineering

Semester: VI

Subject Code & Name: BTCVE605A_Y18 & Waste Water Treatment

Max Marks: 60

Date: 10/07/25

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

	(Level /CO)	Marks
Q. 1 Objective type questions. (Compulsory Question)		12
1 Wastewater treatment mainly aims to:	L1	1
a. Increase water consumption		
b. Remove harmful contaminants		
c. Produce drinking water only		
d. Add nutrients to water		
2 Septic tank is mainly used for:	L1	1
a. Final treatment of water		
b. Preliminary treatment only		
c. Treatment in rural areas		
d. Removal of heavy metals		
3 Oxidation ditch is an example of:	L1	1
a. Primary treatment		
b. Tertiary treatment		
c. Low cost wastewater treatment		
d. Membrane process		
4 The DO sag curve is related to:	L1	1
a. Storm water drainage		
b. Stream pollution and self-purification		
c. Soil contamination		
d. Air pollution control		
5 Neutralization in industrial wastewater treatment is used to:	L1	1
a. Remove oil		
b. Adjust pH		
c. Remove solids		
d. Increase BOD		
6 Phosphorus removal is generally part of:	L1	1
a. Preliminary treatment		
b. Secondary treatment		
c. Advanced treatment		
d. Screening		
7 CETPs stand for:	L1	1
a. Community Effluent Testing Plants		
b. Common Effluent Treatment Plants		
c. Combined Emission Testing Points		
d. Central Effluent Tertiary Plants		
8 Mosquitoes mainly spread:	L1	1
a. Typhoid		
b. Malaria		
c. Ringworm		
d. Tuberculosis		

9	Oxidation pond mainly works on:	L1	1
	a. Membrane filtration b. Anaerobic digestion		
	c. Natural biological processes d. Chemical oxidation		
10	Low cost excreta disposal systems are especially needed in:	L1	1
	a. Urban areas b. Coastal cities		
	c. Rural areas d. Industrial zones		
11	The main difference between sewage and sullage is:	L1	1
	a. Sullage contains storm water b. Sewage does not contain human excreta		
	c. Sullage does not contain human excreta d. Both are same		
12	Ion exchange method is mainly used for removing:	L1	1
	a. Nutrients b. Pathogens		
	c. Heavy metals d. Floating debris		
Q. 2	Solve the following.		12
A)	Write note on various physical properties of waste water.	L1	6
B)	Discuss various preliminary treatment of wastewater.	L2	6
Q.3	Solve the following.		12
A)	Explain the principle of stabilization pond.	L2	6
B)	Explain briefly Oxidation Pond with following points:	L3	6
	a) Working Operation b) Merits and Demerits.		
Q. 4	Solve Any Two of the following.		12
A)	Discuss in brief the biological & chemical method of removal of phosphorous from waste water.	L1	6
B)	Write a short note on membrane separation process.	L2	6
C)	What do you mean by common effluent treatment plant? What is the need of CETP?	L3	6
Q.5	Solve Any Two of the following.		12
A)	Discuss communicable diseases and methods of communication.	L1	6
B)	Explain methods of fly control and engineering approaches to prevent fly breeding.	L2	6
C)	Describe plague control methods and role of rodents in public health.	L3	6
Q. 6	Solve Any Two of the following.		12
A)	Explain in brief essentials of rural sanitation.	L1	6
B)	Outline main features & provisions under swachh Bharat Abhiyan (Clean India Mission).	L2	6
C)	Discuss low cost excreta disposal systems suitable for rural areas.	L3	6

*** End ***